

MACKAY AP, AustraliaWMO: **953670**Lat: **21.1706S**Lon: **149.1794E**Elev: **5**StdP: **101.26**Time Zone: **10.00 (E10)**Period: **96-15**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.1	8.6	2.3	4.5	13.1	4.1	5.1	12.8	10.0	21.4	9.4	21.2	2.8	270	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.9	31.9	25.2	31.0	24.7	30.3	24.4	26.7	29.5	26.1	29.1	25.6	28.7	6.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.1	21.4	28.2	25.3	20.4	28.0	24.7	19.7	27.7	84.2	29.5	81.3	29.1	79.1	28.8	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	10.4	9.5	DB	4.5	34.2	1.7	1.1	3.3	35.0	2.3	35.7	1.4	36.3	0.2	37.1	(4)
(5)			WB	3.3	27.6	1.5	1.1	2.2	28.4	1.3	29.1	0.5	29.7	-0.6	30.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.4	26.8	26.5	25.4	23.2	20.3	18.0	16.8	17.7	20.4	23.2	25.0	26.3	(6)	
	DBStd	4.00	1.31	1.19	1.42	1.57	2.17	2.59	2.48	2.22	2.09	1.90	1.58	1.55	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	148	0	0	0	0	9	37	59	38	6	0	0	0	(9)	
	CDD10.0	4539	519	461	477	397	319	241	212	238	311	409	450	506	(10)	
	CDD18.3	1647	261	228	218	148	70	27	12	18	66	151	200	247	(11)	
	CDH23.3	13144	2664	2226	1818	831	210	39	16	64	341	937	1590	2407	(12)	
	CDH26.7	3043	762	586	361	81	6	2	0	2	36	141	353	714	(13)	
(14) Wind	WSAvg	5.0	5.6	5.7	6.1	5.3	4.8	4.3	4.1	4.2	4.6	5.3	5.2	5.3	(14)	
(15) Precipitation	PrecAvg	1663	296	340	252	154	95	57	34	34	17	47	110	195	(15)	
	PrecMax	2820	1144	1058	832	571	388	367	209	296	75	201	483	1311	(16)	
	PrecMin	853	26	39	16	20	10	3	1	0	0	1	14	14	(17)	
	PrecStd	525	257	249	192	119	70	60	36	50	17	46	98	210	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.7	32.7	31.1	29.8	27.5	25.8	25.0	26.7	29.2	30.6	31.8	33.2	(19)	
		MCWB	26.1	26.3	25.1	23.9	21.6	20.8	18.8	18.9	21.2	23.0	23.8	24.8	(20)	
	2%	DB	31.5	31.1	30.1	28.4	26.1	24.3	23.5	25.0	27.7	29.3	30.4	31.8	(21)	
		MCWB	25.5	25.3	24.5	23.2	21.0	20.2	18.8	19.1	20.9	22.5	23.4	24.7	(22)	
	5%	DB	30.6	30.2	29.4	27.6	25.2	23.5	22.6	23.7	26.4	28.2	29.5	30.8	(23)	
		MCWB	25.0	25.1	24.0	22.7	20.7	19.7	18.2	18.6	20.2	22.1	23.0	24.2	(24)	
	10%	DB	29.9	29.6	28.6	26.9	24.5	22.6	21.8	22.7	25.2	27.2	28.6	30.0	(25)	
		MCWB	24.7	24.9	23.7	22.3	20.4	19.0	17.7	18.3	19.7	21.3	22.4	23.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.8	27.7	26.8	25.7	24.1	22.6	21.6	21.7	23.2	25.2	25.1	27.1	(27)	
		MCDB	30.2	30.8	28.8	27.8	25.5	24.6	23.3	23.9	26.9	27.5	29.8	30.2	(28)	
	2%	WB	26.7	26.7	25.8	24.7	22.8	21.6	20.4	20.7	22.1	24.0	24.4	25.9	(29)	
		MCDB	29.6	29.3	28.5	26.8	24.5	23.4	22.3	23.3	25.6	27.1	28.9	29.6	(30)	
	5%	WB	26.1	26.1	25.2	23.8	21.9	20.8	19.5	20.0	21.5	23.1	23.8	25.2	(31)	
		MCDB	29.2	28.8	27.9	26.4	24.2	22.6	21.5	22.7	24.9	26.9	28.1	29.0	(32)	
	10%	WB	25.5	25.6	24.6	23.1	21.2	20.0	18.8	19.1	20.7	22.4	23.2	24.7	(33)	
		MCDB	28.8	28.5	27.4	25.8	23.6	21.9	20.8	21.9	24.2	26.4	27.4	28.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	5.7	6.2	7.2	8.5	8.8	10.3	10.7	10.0	8.2	7.1	6.8	(35)	
		MCDBR	7.1	7.1	7.1	8.1	8.9	8.7	10.2	11.5	11.6	9.4	8.1	8.2	(36)	
	5% WB	MCWBR	2.8	3.1	3.2	4.2	4.9	5.2	6.0	6.6	5.8	4.1	3.1	2.9	(37)	
		MCWBR	5.9	6.1	5.9	6.9	6.9	6.7	7.5	9.2	9.2	8.1	7.1	7.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.405	0.397	0.385	0.362	0.344	0.332	0.317	0.325	0.355	0.384	0.396	0.400	(40)	
		taud	2.509	2.534	2.552	2.599	2.606	2.630	2.656	2.618	2.522	2.466	2.469	2.492	(41)	
	Ebn at Noon		941	935	922	904	884	877	904	929	936	937	942	946	(42)	
		Edh at Noon	115	110	104	92	85	79	79	89	105	116	119	117	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.32	5.95	5.49	4.87	4.19	3.72	4.21	5.06	6.13	6.77	6.90	6.87	(44)	
	RadStd		0.47	0.65	0.52	0.47	0.27	0.37	0.32	0.35	0.43	0.39	0.80	0.72	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page